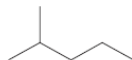
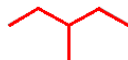
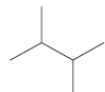
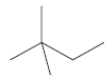
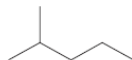
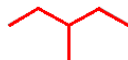
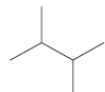
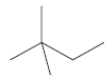
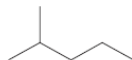
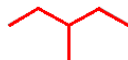
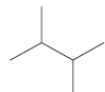
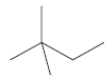


Question				Marking details	Marks available																												
					AO1	AO2	AO3	Total	Maths	Prac																							
7.	(a)	(i)		compounds with the same molecular formula but a different structure / structural formula	1			1																									
		(ii)		<table><tr><th>Name</th><th>Shortened structural formula</th><th>Skeletal formula</th><th>Boiling temperature / °C</th></tr><tr><td>hexane</td><td>CH₃CH₂CH₂CH₂CH₂CH₃</td><td></td><td>69</td></tr><tr><td>2-methylpentane</td><td>CH₃CH(CH₃)CH₂CH₂CH₃</td><td></td><td>62</td></tr><tr><td>3-methylpentane</td><td>CH₃CH₂CH(CH₃)CH₂CH₃</td><td></td><td>63</td></tr><tr><td>2,3-dimethylbutane</td><td>(CH₃)₂CHCH(CH₃)₂</td><td></td><td>58</td></tr><tr><td>2,2-dimethylbutane</td><td>CH₃C(CH₃)₂CH₂CH₃</td><td></td><td>50</td></tr></table> award (1) for each correct answer accept (CH₃)₂CHCH₂CH₂CH₃ for 2-methylpentane	Name	Shortened structural formula	Skeletal formula	Boiling temperature / °C	hexane	CH ₃ CH ₂ CH ₂ CH ₂ CH ₂ CH ₃		69	2-methylpentane	CH ₃ CH(CH ₃)CH ₂ CH ₂ CH ₃		62	3-methylpentane	CH ₃ CH ₂ CH(CH ₃)CH ₂ CH ₃		63	2,3-dimethylbutane	(CH ₃) ₂ CHCH(CH ₃) ₂		58	2,2-dimethylbutane	CH ₃ C(CH ₃) ₂ CH ₂ CH ₃		50		3	3		
Name	Shortened structural formula	Skeletal formula	Boiling temperature / °C																														
hexane	CH ₃ CH ₂ CH ₂ CH ₂ CH ₂ CH ₃		69																														
2-methylpentane	CH ₃ CH(CH ₃)CH ₂ CH ₂ CH ₃		62																														
3-methylpentane	CH ₃ CH ₂ CH(CH ₃)CH ₂ CH ₃		63																														
2,3-dimethylbutane	(CH ₃) ₂ CHCH(CH ₃) ₂		58																														
2,2-dimethylbutane	CH ₃ C(CH ₃) ₂ CH ₂ CH ₃		50																														

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
		(iii)		as the main carbon chain length increases, the boiling temperature increases (or converse) (1) award (1) for either of following lengthening the chain increases the Van der Waals forces between molecules more heat energy is required to overcome the increased forces (or converse)		1	1	2		
	(b)	(i)		$\text{C}_6\text{H}_{14} + 9\frac{1}{2}\text{O}_2 \rightarrow 6\text{CO}_2 + 7\text{H}_2\text{O}$ accept $2\text{C}_6\text{H}_{14} + 19\text{O}_2 \rightarrow 12\text{CO}_2 + 14\text{H}_2\text{O}$ formulae (1) balancing (1) – only awarded if all formulae are correct		2		2		
		(ii)		the number and types of bond are the same (1) the energy absorbed by breaking bonds and the energy released by forming bonds will be similar (1)		2		2		
		(iii)		it has the lowest boiling point / is the most volatile / vaporises most quickly accept explanation of lower boiling point		1		1		
		(iv)	I	energy absorbed = $(5 \times 348) + (14 \times 413) + (6.5 \times 495) = 10739.5$ (1) energy released = $(6 \times 1072) + (14 \times 464) = 12928$ (1) enthalpy change = $10739.5 - 12928 = -2188.5$ (1)		3		3	2	

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
			II	award (2) for quantitative comparison e.g. about twice as much energy is released (with excess oxygen) / additional 1971.5 kJ mol ⁻¹ released (with excess oxygen) award (1) for simple statement e.g. the fuel releases more energy (with excess oxygen) ecf possible from part I			2	2		
			III	award (1) for any reference to CO/carbon monoxide	1			1		
				Question 7 total	2	12	3	17	2	0